

For all calculations – show your work including equations used. You do not need to show units in the calculations but make sure to show your final answer in the correct units (and units do provide a built-in check).

Q1 (5 points)

A sample of unknown gas A (g) escapes through a tiny hole into a vacuum chamber in 187.5 s. The same concentration of unknown gas X₂ (g) escapes through the hole into the evacuated vacuum chamber in 101.3 s under identical conditions. You are given that X₂ (g) is a homonuclear diatomic gas, and that a 13.129 g sample of A (g) in a 1.500 L flask at 25.00 °C has a measured pressure of 1.631 atm. Identify both A (g) and X₂ (g). Justify your answer with calculations. (4 significant figures)

1. [5pts] ? A(g) effuses in 187.5 s $\Delta[A(g)] = \Delta[X_2(g)]$
 (457) ? X₂(g) effuses in 101.3 s

[2pts] $M_A = \frac{mRT}{PV} = \frac{13.129 \text{ g} \cdot 0.082057 \frac{\text{L atm}}{\text{K mol}} \cdot [25.00 + 273.15 \text{ K}]}{1.631 \text{ atm} \cdot 1.500 \text{ L}}$

$M_A = 131.3 \text{ g/mol} \rightarrow A(g) = \text{Xe}(g)$

[3pts] $\frac{R_{X_2}}{R_{Xe}} = \frac{\Delta[X_2]/t_{X_2}}{\Delta[Xe]/t_{Xe}} = \frac{\sqrt{M_{Xe}}}{\sqrt{M_{X_2}}}$

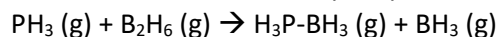
$\frac{t_{Xe}}{t_{X_2}} = \frac{\sqrt{M_{Xe}}}{\sqrt{M_{X_2}}}$ $M_{X_2} = M_{Xe} \left(\frac{t_{X_2}}{t_{Xe}} \right)^2 = 131.3 \frac{\text{g}}{\text{mol}} \left(\frac{101.3 \text{ s}}{187.5 \text{ s}} \right)^2$

$M_{X_2} = 38.39 \frac{\text{g}}{\text{mol}} \quad M_X = 19.1 \frac{\text{g}}{\text{mol}}$

$X_2 = \text{F}_2(g)$ $\leftarrow X = \text{F} \quad (M_F = 19.0 \frac{\text{g}}{\text{mol}})$

Q2 (10 points)**a) 7 points**

The rate of the reaction of phosphine with diborane was measured as a function of temperature.



A plot of $\ln(k)$ versus the reciprocal of absolute temperature obtained from the kinetic data for this reaction had a slope of -5773 K .

At what temperature (in K) would you predict a doubling of the rate for this reaction relative to the rate at room temperature 25.00°C ? Justify your answer showing full calculations. (4 significant figures)

2. A) [7 pts] Plot $\ln k$ vs $1/T$ linear $\rightarrow m = -5773 \text{ K}$ (slope)

(4 sf) $k = A e^{-E_a/RT}$ $\ln k = -\frac{E_a}{R} \left(\frac{1}{T} \right) + \ln A$

[2 pts] $m = -\frac{E_a}{R}$ $E_a = -mR = -(-5773 \text{ K}) \cdot 8.31446 \text{ J K}^{-1} \text{ mol}^{-1}$

$E_a = 4.800 \times 10^4 \text{ J mol}^{-1}$

[5 pts] T_2 where $k_2 = 2 \cdot k_1$ $T_1 = 298.15 \text{ K}$

$\ln \left(\frac{k_2}{k_1} \right) = \frac{E_a}{R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$

$\ln \left(\frac{2k_1}{k_1} \right) = \left[\frac{4.800 \times 10^4 \text{ J mol}^{-1}}{8.31446 \text{ J K}^{-1} \text{ mol}^{-1}} \right] \left[\frac{1}{298.15 \text{ K}} - \frac{1}{T_2} \right]$

$1.201 \times 10^{-4} \text{ K}^{-1} = 3.3540 \times 10^{-3} \text{ K}^{-1} - \frac{1}{T_2}$

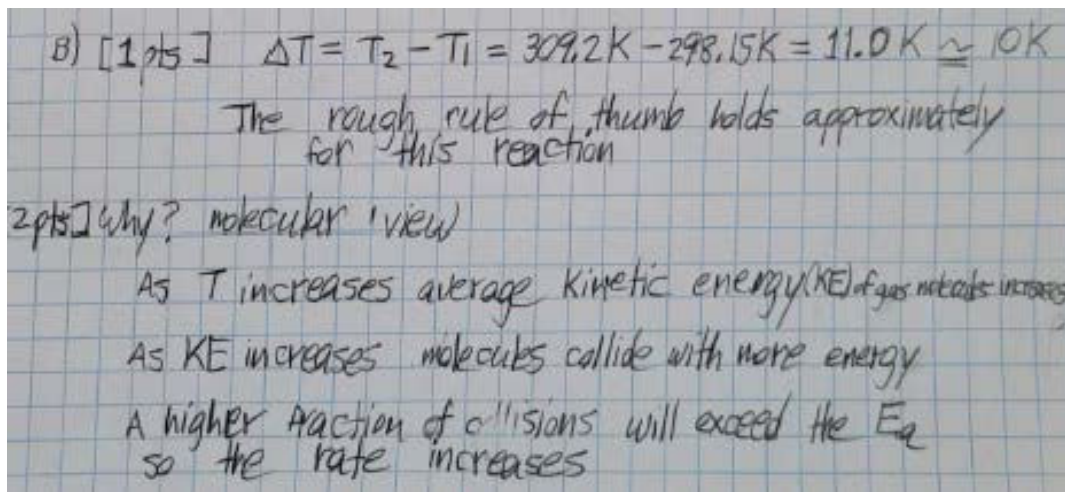
$-3.234 \times 10^{-3} \text{ K}^{-1} = -\frac{1}{T_2}$

$T_2 = 309.2 \text{ K}$

b) 3 points

There is a rough "rule of thumb" in chemical kinetics that predicts a doubling of the rate of a reaction for every 10 degrees increase in temperature.

Does this approximate kinetic rule of thumb estimate hold for the reaction in part a)? Briefly provide a physical chemical justification in terms of gas molecular behavior for this approximate kinetic temperature relationship.



B) [1 pt] $\Delta T = T_2 - T_1 = 309.2\text{K} - 298.15\text{K} = 11.0\text{K} \approx 10\text{K}$

The rough rule of thumb holds approximately for this reaction

2 pts] why? molecular view

As T increases average kinetic energy (KE) of gas molecules increases

As KE increases molecules collide with more energy

A higher fraction of collisions will exceed the E_a so the rate increases

Q3 (10 points)

A solution of a newly discovered polymer is prepared by dissolving 30.00 milligrams of polymer in 1.00 millilitre of water. This sample exerts an osmotic pressure of 1239.5 Pascal at 25°C.

What is the molecular weight of the polymer? (3 significant figures)

$$1239.5 \text{ Pa} = 1.2395 \text{ kPa}$$

$$\Pi = MRT$$

$$1.2395 \text{ kPa} = M \cdot 8.3145 \text{ kPa}\cdot\text{L}/\text{mol}\cdot\text{K} \cdot 298.15 \text{ K}$$

$$M = 1.2395 \text{ kPa} / (8.3145 \text{ kPa}\cdot\text{L}/\text{mol}\cdot\text{K} \cdot 298.15 \text{ K})$$

$$M = 5.0001 \cdot 10^{-4} \text{ mol/L}$$

Which is $5 \cdot 10^{-4}$ mol in 1 L.

There are several ways to proceed, here's one:

30 mg/mL is the same as 30 grams in 1 L.

So 30 grams is $5 \cdot 10^{-4}$ mol so:

$$\text{M.W.} = 30 \text{ grams} / 5 \cdot 10^{-4} \text{ mol} = 6.00 \cdot 10^4 \text{ gr/mol}$$

Q4) 10 points

A new element has been discovered, Sewallium (Sw)! It forms an ionic compound with chlorine that acts as a strong electrolyte in water. You are in charge of determining the number of chloride ions bound to each Sewallium ion, using freezing point depression.

In a carefully controlled experiment a 1.00×10^{-3} molal solution of Sewallium Chloride in water has a freezing point depression of 0.00735 K.

Using this measurement determine the most likely chemical formula for Sewallium Chloride from among the following options: SwCl, SwCl₂, SwCl₃, SwCl₄. Justify your answer showing relevant calculations.

Expected $\Delta T_f = 0.001 \text{ mol/kg} * -1.86 \text{ C}/(\text{mol/kg}) = -0.001860 \text{ C}$ (same in K)

This leads to a Van't Hoff Factor calculation that will show how many particles

$i = 0.00735 \text{ K} / -0.001860 \text{ K} = 3.95$

So dissolving Sewallium Chloride in water produces 4 particles per mole, implying SwCl₃ is the most probable formula:



Q5) 6 points

A sample is prepared by adding 7271.7 grams of ethylene glycol to 69308.9 grams of water, which forms a solution that has a total volume of 75.68 liters.

The molecular weight of ethylene glycol is 62.07 gr/mol.

The molecular weight of water is 18.02 gr/mol.

Calculate the concentration of ethylene glycol in this solution expressed as (3 significant figures for each):

a. mole fraction

b. molality

c. molarity

7271.7 grams EG with M.W. 62.07 gr/mol , is 117.1532141 moles

69308.9 grams (69.3089 kg) water with M.W. 18.02 gr/mol so 3846.220866 moles.

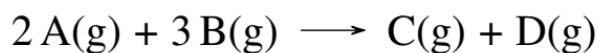
Mole Fraction : $117.15 \text{ mol} / (117.15 + 3846.22) \text{ mol} = 0.0296$

Molarity = $117.15 \text{ mol} / (75.68 \text{ L}) = 1.55 \text{ mol/L}$

Molality = $117.15 \text{ mol} / 69.3089 \text{ kg} = 1.69 \text{ mol/kg}$

Q6) 10 points

Consider the following reaction, where $K_c = 1.57 \times 10^{-26}$:



Calculate the equilibrium concentrations of A, B, C, D starting with 0.30 mol A and 0.60 mol of B combined in total volume 1.00 L. Note that none of the concentrations can be zero at equilibrium. (3 significant figures)

This is a very small equilibrium constant, so the reaction will barely proceed. The concentrations of A and B will therefore be unchanged:

	A	B		C	D
I	0.3	0.6		0	0
C	-2x	-3x		+x	+x
E	0.3 - x ~ 0.3	0.6 - x ~ 0.6		x	x

So we can then use the equilibrium expression

$$K = [\text{C}][\text{D}]/[\text{A}]^2[\text{B}]^3 = x^2 / [0.3]^2[0.6]^3$$

$$K * [0.3]^2[0.6]^3 = 3.05208 \times 10^{-28} = x^2$$

$$x = 1.75 \times 10^{-14} = [\text{C}] = [\text{D}]$$

Q7) 5 points

For the reaction



$$K = 1.87 \times 10^{-7}$$

$$\Delta S^0 = 181.92 \text{ J/(mol}\cdot\text{K)}$$

- a. What is the value of ΔH^0 at 298 K? (3 significant figures)

We can find the value of ΔG^0 from K , and then use that and the value of ΔS^0 in the problem to find ΔH^0 .

$$\Delta G^0 = -8.3145 \text{ J/molK} \times 298 \text{ K} \ln(1.87 \times 10^{-7})$$

$$= 38385.105 \text{ J/mol}$$

$$\Delta G^0 = \Delta H^0 - T \Delta S^0$$

$$\Delta G^0 + T \Delta S^0 = \Delta H^0 = 38385.105 \text{ J/mol} + 298 \text{ K} \times 181.92 \text{ J/(mol}\cdot\text{K)}$$

$$= 92597.265 \text{ J/mol} = 92.6 \text{ kJ/mol or } 92.6 \times 10^3 \text{ J/mol}$$

- b. At what temperature would a mixture containing an equal number of moles of A and B be at equilibrium? (3 significant figures)

We are looking for a temperature where the concentrations of A and B are equal, so $K = 1$ and $\Delta G^0 = 0$.

We can find this using the van't Hoff equation OR the free energy expression, they will give the same answer.

$$\Delta G^0 = \Delta H^0 - T \Delta S^0$$

$$0 = \Delta H^0 - T \Delta S^0 \text{ which is true when } \Delta H^0 = T \Delta S^0$$

We have the enthalpy and entropy, so we can find the temperature

$$\Delta H^0 / \Delta S^0 = T$$

$$92597.265 \text{ J/mol} / 181.92 \text{ J/(mol}\cdot\text{K)} = 509 \text{ K}$$

Q8) 12 points**a) 4 points**

K_a of HX is 4.65×10^{-5} . What is the pH of a 0.500 M solution of NaX at 298 K? (3 significant figures)

$$K_a \text{ of HX} = 4.65 \times 10^{-5} \quad \text{So } K_b \text{ of X}^- = 2.15 \times 10^{-10}$$

$$\text{pOH} = (K_b \times [B])^{0.5} \quad \text{pOH} = 4.984$$

$$\text{pH} = 9.01 \text{ or } 9.02 \text{ (also accepted)}$$

b) 8 points

Arrange the following from lowest to highest pH, using the information given in the table. Anions are bolded. Explain your answer.

0.1 M HBCl; 0.1 M HAX; 0.1 M HBX

Ion or Molecule	$\text{p}K_a$	$\text{p}K_b$
AH^+	8.61	
HX	5.39	
B		3.20

Can show calculations but do not need to do calculations. When calculating, have to account for the conjugate base and acid. For example: HBX – would need to calculate pH not just based on HB^+ but also X^- . This calculation is tricky to do, and is not needed to solve the problem

All are same concentration:

HBCl : HB^+ and Cl^- . Cl^- has no effect on pH. HB^+ is the cation which is acidic (conjugate of weak base B). So 0.1 M HBCl is acidic

HAX: HA^+ ($\text{p}K_a = 8.61$) and X^- ($\text{p}K_b = 8.61$; conjugate base of HX). Since $\text{p}K_a = \text{p}K_b$. This solution is neutral

HBX: HB^+ and X^- . $\text{p}K_a$ of HB^+ is 10.80 and $\text{p}K_b$ of X^- is 8.61. Since $\text{p}K_b$ of X^- is lower, this is a stronger base.

Order: 0.1 M HBCl, 0.1 M HAX, 0.1 M HBX

Q9) 15 points**a) 4 points**

In a chemistry experiment, you must choose possible reagents to carry out a reaction to form the highly reactive anionic base, X^- , from its conjugate acid (pK_a of the acid, HX, is 25).

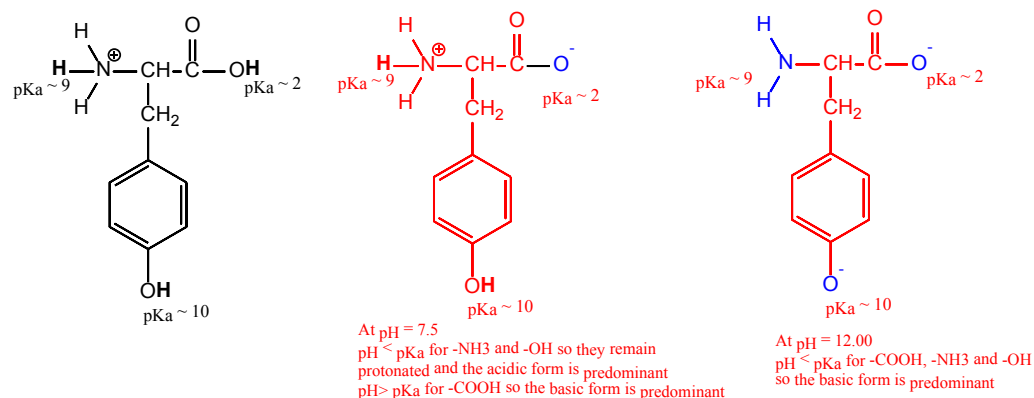
Using the data below, identify the reagent(s) you could use to carry out the reaction? You can choose more than one reagent. You must provide the reagent(s) in its correct form (acid or the conjugate base) and justify your answer. You do not need to provide counter-cation or anion.

Acid	pK_a
HA	10
HB	20
HC	35
HD	50

Can be C- or D- (must provide the anionic conjugate base of acids that are weaker than HX. Weaker acid = stronger conjugate base).

b) 6 points

Draw the most prominent ionization state for the following compound at (i) pH = 7.50 and (ii) pH = 12.00, at 298 K using the data given below. The pK_a values given are for the hydrogens shown in bold. Justify your answer.



c) 5 points

K_a for HX is 2.5×10^{-8} , calculate the equilibrium constant (K) value for the reaction of $\text{HX}_{(\text{aq})}$ with $\text{OH}^-_{(\text{aq})}$ at 298 K. Show relevant equations and calculations. (3 significant figures)

$$K_b \text{ } X^- = \frac{K_w}{K_a} = \frac{10^{-14}}{2.5 \times 10^{-8}} = 4.00 \times 10^{-7}$$

$$K_b : \quad X^-_{\text{aq}} + \text{H}_2\text{O}_{\text{l}} \rightleftharpoons \text{HX}_{\text{aq}} + \text{OH}^-_{\text{aq}}$$

$$K_b = \frac{[\text{HX}][\text{OH}^-]}{[X^-]} \quad \frac{[X^-]}{[\text{HX}]} = \frac{[\text{OH}^-]}{K_b}$$

CALCULATE K :

$$\text{HX}_{(\text{aq})} + \text{OH}^-_{(\text{aq})} \rightleftharpoons X^-_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})}$$

$$K = \frac{[X^-]}{[\text{HX}][\text{OH}^-]} = \frac{[\text{OH}^-]}{K_b [\text{OH}^-]} = \frac{1}{K_b} = \boxed{2.50 \times 10^6}$$

The reaction is the reverse of the reaction that describes the K_b of X^-

Q10) 6 points

For a polyprotic acid, H_2A :

$$K_{a,1} = 1.50 \times 10^{-3}$$

$$K_{a,2} = 2.50 \times 10^{-11}$$

You need to conduct an important kinetic assay at $pH = 10.5$. Using the data above, choose a buffer system (acid/conjugate base) that can be used for your assay. Justify your answer.

What is the ratio of acid to conjugate base at $pH = 10.5$? Show calculations.

$$pK_{a,1} \text{ between } H_2A/HA^- = -\log(K_{a,1}) = 2.82$$

$$pK_{a,2} \text{ between } HA^-/A^{2-} = -\log(K_{a,2}) = 10.60$$

Choose buffer system that is ± 1 within the pK_a

So a buffer with acid HA^- and base A^{2-} would be the correct choice.

Use Henderson Hasselbalch equation to determine the ratio of acid/base:

$$pH = pK_a + \log [base]/[acid] \text{ (can use any form of this equation)}$$

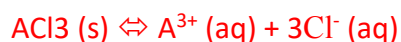
$$-0.1 = \log [base]/[acid]$$

$$[Acid]/[Base] = 1.26: 1$$

Q11) 5 points

In one of their missions for the starship Enterprise, Spock and Sulu discover a new element, A. They calculate the solubility of the ionic salt, ACl_3 , in water to be 5.87 mg/L. What is the K_{sp} of ACl_3 ? Molecular weight of ACl_3 is 393 g/mol. Show your work and calculations. (3 significant figures)

$$\text{Solubility} = 5.87 \times 10^{-3} \text{ g/L} = 1.4936 \times 10^{-5} \text{ mol L}^{-1}$$



	ACl_3	A^{3+}	Cl^{-}
Initial		-	-
Change		+x	+3x
Equilibrium		X	3x

For every mole of ACl_3 there is 1 mole of A^{3+} and 1 mole of Cl^{-}

Since solubility of $\text{ACl}_3 = 1.4936 \times 10^{-5} \text{ mol L}^{-1}$

$$X = 1.4936 \times 10^{-5} \text{ mol L}^{-1}$$

$$K_{\text{sp}} = [\text{A}^{3+}] [\text{Cl}^{-}]^3 = (x) (3x)^3$$

$$K_{\text{sp}} = 27x^4$$

$$K_{\text{sp}} = 1.34 \times 10^{-18} \text{ (1.33} \times 10^{-18} \text{ also acceptable)}$$

Q12) 16 points

20.00 mL of a 0.1000 M solution of a weak base ($K_b = 2.2 \times 10^{-5}$) is titrated with 0.1500 M strong acid, HCl, at 298K.

i. Calculate the initial pH of the solution

$$\text{Initial } [\text{OH}^-] = (0.1 \times K_b)^{0.5} = 1.483 \times 10^{-3}$$

$$\text{pOH} = 2.82879; \text{pH} = 11.17 \text{ or } 11.2 \text{ (Can show ICE table but not needed)}$$

ii. Calculate the pH of the solution after addition of 5.00 mL of HCl. Show your calculations (3 significant figures)

After 5.00 mL of 0.15M acid is added (Can show ICE table for below)

$$\text{Initial moles of base} = 2.00 \times 10^{-3} \text{ mol}$$

$$\text{Moles of strong acid added} = 7.5 \times 10^{-4} \text{ mol}$$

$$\text{So moles of base remaining} = 1.25 \times 10^{-3} \text{ mol}$$

$$\text{Mole of conjugate acid} = 7.5 \times 10^{-4} \text{ mol}$$

Can convert the above moles to concentration but do not need to as the volumes are same (both using moles or concentration are acceptable)

$$\text{pH} = \text{p}K_a + \log [\text{base}]/[\text{acid}]$$

$$\text{p}K_a = K_w/\text{p}K_b = 4.545 \times 10^{-10}$$

$$\text{pH} = 9.342 + \log (0.00125/0.00075) = 9.56$$

iii. Calculate the pH of the solution after addition of 30.00 mL of HCl. Show your calculations (3 significant figures)

Beyond equivalence point. Calculate excess acid

$$\text{Moles of strong acid added} = 4.50 \times 10^{-3} \text{ mol}$$

$$\text{Excess of moles of acid} = 4.50 \times 10^{-3} - 2.00 \times 10^{-3} = 2.50 \times 10^{-3} \text{ moles}$$

$$[\text{H}_3\text{O}^+] = 2.50 \times 10^{-3} \text{ mol} / 50.00 \times 10^{-3} \text{ L} = 0.05 \text{ M}$$

$$\text{pH} = -\log (0.05) = 1.30$$

iv. Which of the following indicator(s) would you choose for this titration? Justify your choice with calculations.

Indicator	$pK_a(s)$	Indicator	$pK_a(s)$
Thymol Blue	1.65 and 8.9	Methyl Yellow	3.3
Bromophenol Blue	3.85	Methyl Red	4.95
Phenolphthalein	9.4	Alizarin Yellow	11.2

Calculate pH at equivalence point

Moles of base = moles acid = 2.00×10^{-3}

Macid Vacid = 2.00×10^{-3} (Macid added = 0.15 M)

Volume of acid at equivalence = 0.0133333 L

[HA] at equivalence = $0.0020/(0.0333) = 0.06$ M

Acid is in excess (weak base strong acid titration)

$HA + H_2O \rightleftharpoons A^- + H_3O^+$

	HA	H ₂ O	A ⁻	H ₃ O ⁺
Initial	0.06	-		
Change	-x	-	x	x
Equilibrium	0.060-x	-	x	x

$K_a = 4.545 \times 10^{-10} = x^2/(0.06-x)$

Solve for x (do not need to solve a quadratic but will obtain the same answer if used a quadratic)

$x = 5.22 \times 10^{-6}$

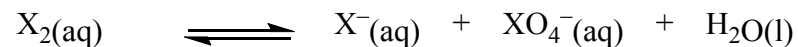
pH = 5.28 at equivalence

So use methyl red as the indicator (+/- 1 within the pK_a of the indicator)

Q13) 10 points**a) 4 points**

Balance the following equation using smallest whole number coefficients. Show half-reactions and the final fully balanced equation using smallest whole number coefficients.

In basic media



(X is a non-metal and is less electronegative than O)

Oxidation half reaction: $\text{X}_2 + 8\text{H}_2\text{O} \rightleftharpoons 2\text{XO}_4^- + 16\text{H}^+ + 14\text{e}^-$

Reduction half reaction: $(\text{X}_2 + 2\text{e}^- \rightleftharpoons 2\text{X}^-) \times 7$

Overall reaction



Lowest coefficient

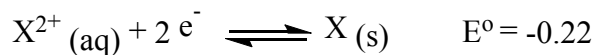


In basic media so add 14 OH⁻ to either side



b) 6 points

A voltaic cell is constructed using the reagents based on the following half reactions:



Calculate the initial voltage (2 significant figures) generated by the cell at 25 °C, if the initial concentration of $\text{X}^{2+}(\text{aq}) = 0.20 \text{ M}$; $\text{Y}^{2+}(\text{aq}) = 0.25 \text{ M}$; and $\text{Y}^{3+}(\text{aq})$ is 0.0020 M .

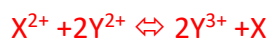
Show your work, including any equations used, and calculations.

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{RT}{zF} \ln Q$$

$$R = 8.314 \text{ J/mol K}; F = 96485$$

X^{2+} will be reduced and Y^{2+} will be oxidized

Overall reaction:



So $z = 2$

$$E_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}} = -0.22 - (-0.51) = 0.29 \text{ V}$$

$$Q = \frac{[\text{Y}^{3+}]^2}{[\text{Y}^{2+}]^2[\text{X}^{2+}]}$$

Answer does not change but the equation for Q given originally was incorrect:

$$Q = \frac{[\text{Y}^{3+}]^2}{[\text{Y}^{2+}]^2[\text{X}^{2+}]}$$

Solve for E_{cell}

$$= 0.290 \text{ V} + 0.103 \text{ V} = 0.393 \text{ V} \text{ (0.39 V accepted)}$$

Q14) 12 points**a. 9 points**

Some batteries consist of the Zn (s) and MnO₂ (s) electrodes. At the Zn electrode ZnO(s) forms, and at the MnO₂ electrode Mn₂O₃(s) forms.

Using the above information, write the half reactions for both anode and cathode, and the balanced overall reaction. Show your work.

Oxidation (Anode) half reaction:



Reduction (cathode) half reaction:



Overall reaction (+2):



Calculate the mass in kg (3 significant figures) of ZnO (s) formed if 1.0×10^4 A are passed through the cell for 10 hours. Show your work.

$$\text{Current} = 10^4 \text{ A}$$

$$\text{Time} = 10 \times 60 \times 60 = 36,000 \text{ seconds}$$

$$\text{Charge} = 3.6 \times 10^8 \text{ C}$$

$$\text{Number of moles of electrons} = 3.6 \times 10^8 / F = 3731.15 \text{ mol}$$

$$\text{Number moles of ZnO produced} = 3731.15 / 2$$

$$\begin{aligned} \text{Mass of ZnO produced} &= \text{moles of ZnO} \times \text{molar mass of ZnO} \\ &= 152 \text{ kg} \end{aligned}$$

b. 3 points

What are the electrolysis products for the following mixture? Justify your answer.

$\text{AlCl}_3 + \text{KBr}$ (molten)

At cathode compare cations: Al^{3+} or K^+

Al^{3+} has higher ionization energy so at cathode Al^{3+} is reduced to Al(s)

At anode compare anions: Cl^- or Br^-

Br^- is less electronegative so Br^- will be oxidized to Br_2 at the anode

Products Al(s) and $\text{Br}_2(\text{l})$

Q15) 12 points

Using the table given below, answer the following questions:

Half reactions	E° (volts)
$A^+ (aq) + e^- \rightarrow A(s)$	1.73
$B^{2+} (aq) + 2e^- \rightarrow B(s)$	-0.14
$C^{2+} (aq) + 2e^- \rightarrow C(s)$	-0.41
$X_2 (g) + 2e^- \rightarrow 2X^- (aq)$	1.28
$D^{2+} (aq) + 2e^- \rightarrow D(s)$	-0.17
$E^{2+} (aq) + e^- \rightarrow E^+(aq)$	-0.29

a) 2 points

Which reagent is the strongest reducing agent? Justify your choice briefly.

C(s) is the strongest reducing agent (most negative reduction potential of C^+ , so C^+ is the weakest oxidizing agent, and hence C is the strongest reducing agent)

b) 4 points

Which reagent(s) will reduce $B^{2+} (aq)$ while leaving C^{2+} unreacted? Justify your choice briefly.

D(s) and $E^+ (aq)$ will reduce B^{2+} as the reduction potential is lower than B^{2+} but higher than C^{2+}

c) 6 points

You want to obtain a voltaic cell that has the highest cell potential from the data given in the table. Write down the overall balanced equation and the E° cell value for the cell with the highest potential. Justify your choice.

The highest E° cell value will be obtained from the combination of the two half reactions with the maximum and minimum reduction potential.

Reduction reaction: $A^+ + e^- \rightleftharpoons A$

Oxidation reaction: $C \rightleftharpoons C^{2+} + 2e^-$

Overall reaction:

$2A^+ (aq) + C (s) \rightleftharpoons C^{2+} (aq) + 2A (s)$

$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$
 $= 1.73 - (-0.41) = 2.14 \text{ V}$